

Runzhe Yang

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🏠 Homepage

Personal Details

Date of Birth December 31st, 2001
Citizenship Chinese
Gender Male

Educations

- 10/2026 – **Ph.D. in Math (Mathematical Finance)**, Department of Mathematics, Imperial College
Current London, London, England.
Advisors: Prof. Dr. Antonie Jacquier & Eyal Neuman.
- 10/2024 – **M.Sc./Phase I in Math (Stochastics & Finance)**, Berlin Mathematical School & Technische
6/2026 Universität Berlin, Berlin, German.
Thesis Title: Second-order Weak Schemes and Error Bounds for Rough Heston and Quadratic
Rough Heston Models.
Advisors: Prof. Dr. Peter Friz & Christian Bayer.
Final Grades: 1.0/1.0.
- 3/2024 – **Visiting Student**, Department of Mathematics, The Hong Kong University of Science and
5/2024 Technology, Hong Kong SAR.
Host: Prof. Dr. Maximilian Nitzschner.
- 9/2020 – **B.Sc. in Math (Probability & Statistics)**, School of Mathematics and Statistics, Wuhan
6/2024 University, Wuhan, Hubei, China.
Thesis Title: Moderate Deviation of Gaussian Fluctuations for Coulomb Gases at Any Temperature.
Advisor: Prof. Dr. Fuqing Gao.
Final Grades: 89.6/100.
- 9/2017 – **Abitur**, Huizhou No.1 Middle School, Huizhou, Guangdong, China.
6/2020 Final Grade: Top 0.3% Provincewide.

Research Interests

My primary research interests lie in the area involving Probability and Analysis, and their applications in Mathematical Finance.

- Mathematical Finance, Rough Volatility and Computational Finance
- Stochastic Analysis and Stochastic (Partial) Differential Equations
- Rough analysis and its applications

Honors and Awards

- 6/2026 Laureate, Roth Scholarship, Department of Mathematics, Imperial College London, 2026 - 2030.
- 6/2024 Laureate, Top 3%, Outstanding Bachelor's Thesis Award, Wuhan University.
- 5/2024 Merit Undergraduate, Top 20%, Wuhan University.
- 2/2024 Laureate, Phase I Scholarship for Student at Berlin Mathematical School (1,000 Euros/Month),
2024 - 2026.
- 1/2024 Laureate, Scholarship for Visiting Internship Student for Research at HKUST (15,000 HKD/Month),
Spring Semester.
- 11/2022 First Class, Top 3%, WHU Outstanding Mathematics Student Scholarship, Wuhan University,
2021-2022 Academic Year.

- 7/2022 Winner's Prize, Rank 19th Nationwide, *the 13th S.-T. Yau College Student Mathematics Contest, Statistics and Probability track (individual)*.
- 12/2021 First Prize (Finalist), Rank 16th Provincewide, *the 13th China National Mathematics Competition for College Students, Wuhan, China*.

Research Projects

4/2025 – **On the Rough and Quadratic Rough Heston Model,**

12/2025 advisor: *Dr. Christian Bayer*.

In this project, we mainly investigate the Rough and Quadratic Rough Heston Model. We proved that the weak scheme introduced by Bayer and Breneis indeed achieves the second-order weak rate. We've also extended the PPDE method by developing Malliavin calculus framework for a broader class of SVEs, which allows us to obtain the path-dependent Feynman–Kac equation and the functional Itô formula, and thereby derived the error bound for the Quadratic Rough Heston model. Besides, an efficient weak scheme for simulating the quadratic rough Heston model is also proposed.

6/2023 – **Lower Bound for Disconnection of Vacant Set in Loop Percolation,**

5/2024 advisor: *Prof. Dr. Maximilian Nitzschner*.

The principal aim of this project is to derive asymptotic lower on the probability that in a certain model of random walk loops on the integer lattice, a macroscopic set is disconnected from infinity by the trace of the loops. We further study the large deviations of the occupation-time field for the loop soup and characterize its supercritical behavior.

Academic Experiences

4/2026 **ViennaSPDEs2026: From discrete systems to SPDEs, TU Wien, Vienna, Austria,**
funded by BMS Travel Grant.

7/2023 **Summer School Series on Probability and Related Fields: Masterclass in Random Geometry, Peking University, Beijing, China,**
funded by Beijing International Center For Mathematical Research (BICMR).

Technical Proficiencies

Computer C/C++, R, L^AT_EX

Language Mandarin/Cantonese (Native), English (Fluent), Germany (Beginner)